

Listening Playground



**Listening and Sound Mapping activities for
young people**

Digital Playgrounds for Music

Listening Playground: About

Listening Playground is part of the Digital Playgrounds for Music project (DPfM), an initiative exploring how digital tools can support playful, creative, and inclusive music learning in schools.

This activity was developed through collaboration between the University of Huddersfield, the University of Cambridge, and University of Cambridge Primary School (UCPS), and has been tested with Year 6 pupils as well as in SEND learning contexts.

DPfM supports the idea that listening can be a creative act that connects us to learning about our world. The Listening Playground app allows young people to explore their everyday environment through sound, science, and technology: noticing, recording, and reflecting on the sonic qualities of places they already know. This document outlines an easy to use lesson plan and app instructions to allow young people to:

- see sound visually using sonograms,
- experiment with music technology principles,
- practice developing an archive through naming and tagging,
- and place sounds onto a shared map to create a collective 'sound portrait' of their school/community.

Curriculum links:

- Expressive Arts: creative listening, sound-making, composition, and interpretation.
- Science: sound as vibration, volume, pitch, and observation of physical phenomena.
- Technology: digital recording, data organisation, tagging.
- Literacy: descriptive language, reflection, discussion, and meaning-making.

Learning goals...By the end of the lesson, students will:

- Understand that sound is vibration and can be explored visually,
- Use microphones and gain to record sounds at different volumes,
- Record, name, and tag sounds,
- Explore their environment through careful listening,
- Reflect on what sound reveals about place and community.



The Listening Playground app is free to use, runs in a web browser on tablets, phones, or computers, and is designed to be flexible. Teachers may use the full lesson as outlined here, adapt individual sections, or return to the activity over time to build a longer-term sound-mapping project.

Accessing the App

<https://mariaguerrasappho.github.io/ListeningPlayground/>

Demonstration video showing the app setup and a full classroom session is also available:

<https://huddersfield.box.com/s/bhzturcucajrd00ovbs2q3bhparu7wef>



About Digital Playgrounds for Music

Digital Playgrounds for Music (DPfM) is a initiative based at the University of Huddersfield exploring how creative technologies invite exploration, curiosity, and agency, and in doing so can deepen musical understanding and enrich learning across subjects.

DPfM treats digital tools as playgrounds: spaces for experimentation, listening, making, and reflection.

- Playful exploration
- Listening and embodied experience as foundations for creativity
- Collaboration and shared meaning-making
- emphasis on inclusive and sensory approaches to music learning.

The activity presented here has been tested with Year 6 pupils and in SEND learning contexts, with observations informing both the design of the software and the structure of the lesson materials included here.

DPfM Software Ecosystem:

Sound Explorer is not an isolated app. It is part of a longer research trajectory exploring how musical ways of learning, supported by creative technology, can enrich learning across subjects.

DPfM tools support a commitment to 'interactive aural analysis': engaging with music as sound itself. The Sound Explorer App represents one example of the broader approaches across the softwares - translating approaches from the wider softwares into a tablet friendly and primary classroom focused activity.

A Brief Overview of DPfM Software:

TIAALS (Tools for Interactive Aural Analysis): A toolbox of approaches that enable musicians, researchers, audiences and students to explore music through listening, interaction, and visualisation.

TaCEM (Technology and Creativity in Electronic Music): Interactive software exploring the relationship between technology and creativity in electronic and electroacoustic music through listening and exploration.

Evolution and Collaboration: An interactive environment exploring collaborative composition and rehearsal in contemporary music, supporting non-linear listening and reflection on creative processes.

<https://research.hud.ac.uk/institutes-centres/projects/digitalplaygroundsformusic/>



Lesson Plan: Listening Playground App

Setting up the App (teachers only)

Duration: 90 minutes

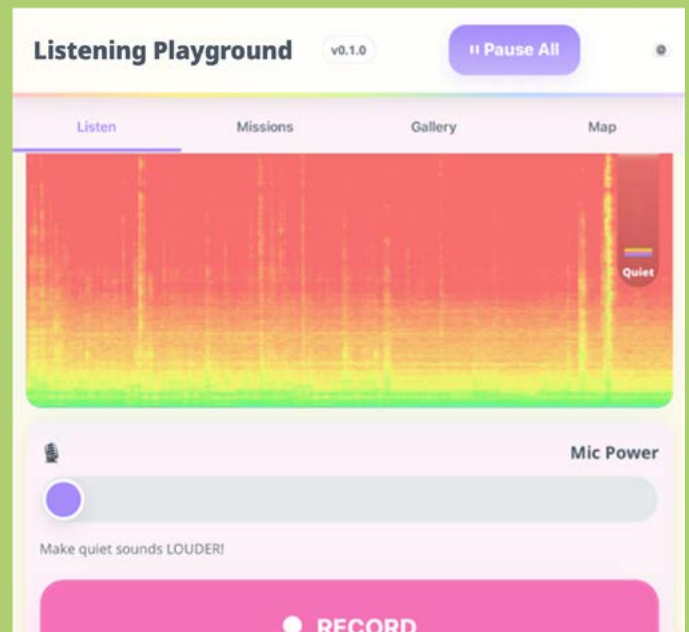
Recommended age range: This lesson was developed and tested with Year 6 students

Access the App: <https://mariaguerrasappho.github.io/ListeningPlayground/>

Video demonstration (recommended):
<https://huddersfield.app.box.com/file/2120086396334?s=bhzturcucajrd00o-vbs2q3bhparu7wef>

Tools and Materials:

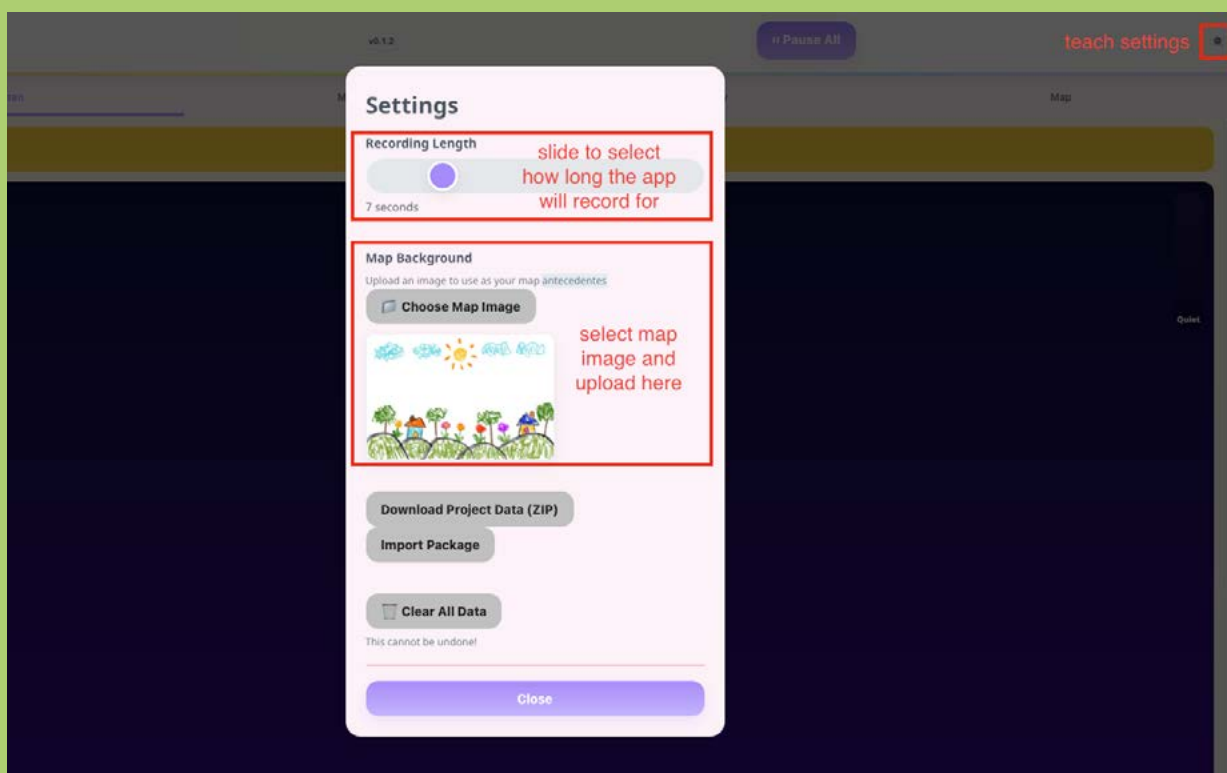
- Enough tablets for students working in small groups
- Internet connection (app needs to be loaded in internet but will work without internet after this)
- Safe outdoor or indoor recording areas



Before the Lesson: Teacher Setup

Please refer to the video demonstration for in depth setup instructions

- Open the Sound Explorer app link
- Click Start Listening and grant access to the microphone and camera
- In Settings [top right corner] upload a background image for the sound map.
- Choosing or creating the map image can be a short activity in itself, or prepared in advance i.e:
 1. simple drawing made by the class, or
 2. bird's-eye photograph or any other photo of the school or learning space
- Check or adjust the recording length (default 7 seconds)
- Pre-select the small groups for students to work in with tablets/devices



Lesson Plan: Listening Playground App

Seeing Sound: What Is a Sonogram? (10-15 minutes)

Opens the app and click **'Start Listening'** and ensure that you accept the app in accessing the devices camera and microphone. Some devices may require further settings to ensure the browser is granted microphone and camera permissions.

Once successfully started the sonogram should begin to roll across the screen. Makes a range of sounds (clap, tap, voice, silence) to demonstrate how the sound around the tablet changes the visuals on the sonogram.

Teacher explains:

"This moving picture shows sound. It's called a sonogram. Allowing us to see sound as it happens."

Key ideas:

Colour shows how loud or energetic a sound is

Low sounds appear near the bottom

High sounds appear near the top

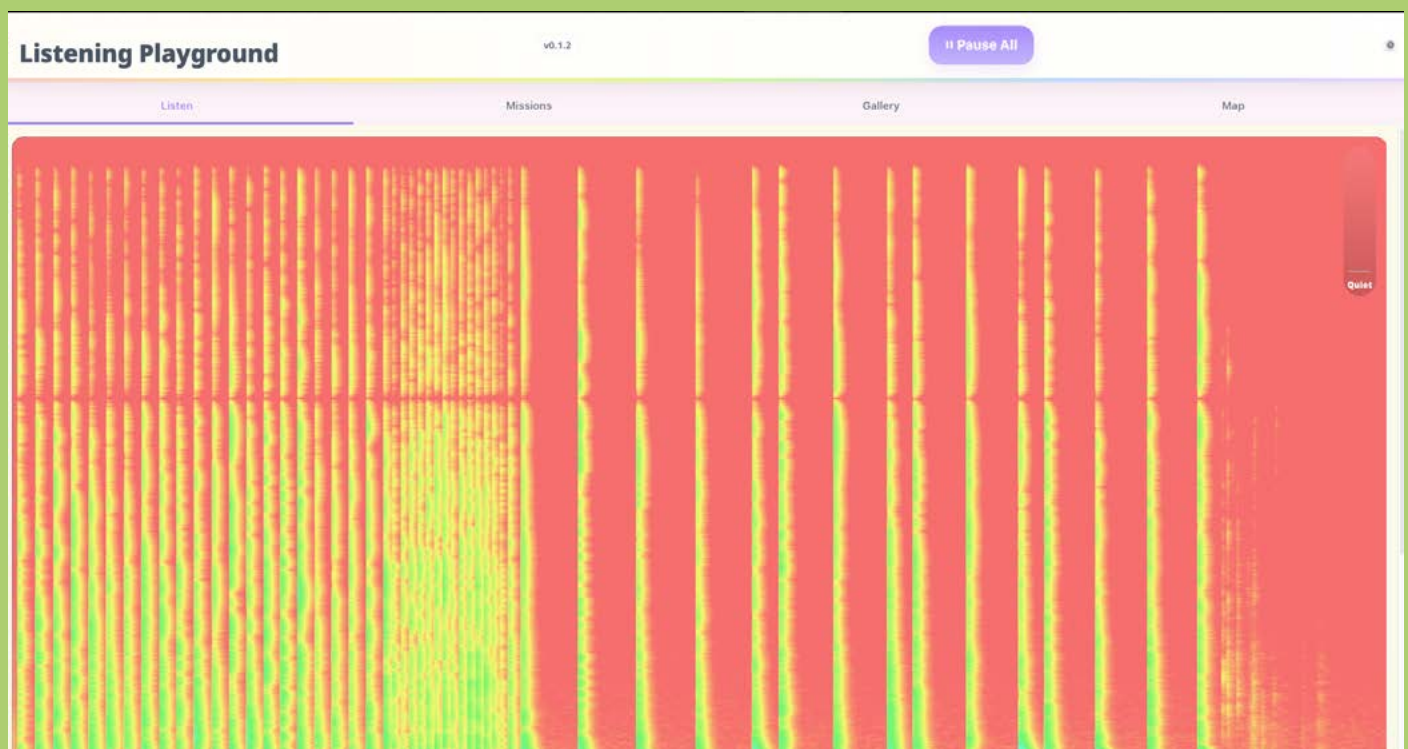
Sound scrolls across the page as time passes

Student prompts:

"What happens when the sound gets louder?"

"What do you notice when we stop making sound?"

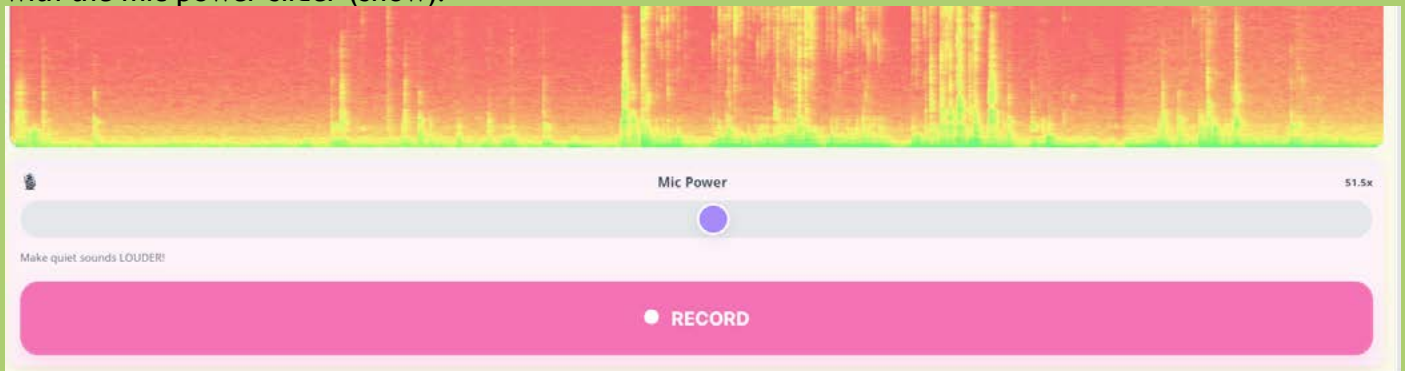
"Can we clap our hands together and make a pattern on the screen? How does it change if we clap slow, or clap fast?"



Lesson Plan: Listening Playground App

Capturing Sound: Microphones and Gain (10 minutes)

Explain: The tablet has a microphone. Gain controls how strongly the microphone listens which we can set with the mic power slider (show).



“Microphone gain is a bit like how we help our ears in different situations.

If someone finds it hard to hear, they might use a hearing aid, which makes quiet sounds louder.

If a place is very noisy, we might wear earplugs to make loud sounds quieter.

Gain works in a similar way for the microphone.

When a sound is very quiet, we turn the gain up so the microphone can hear it clearly.

When a sound is very loud, we turn the gain down so the recording doesn’t become too strong or distorted.”

Activity:

All clap together a few times with low gain: what do we see on screen?

All clap together a few times with high gain: how does it change?

Student prompts:

“What kind of gain would we set for a car honking?”

“What kind of gain would we set for a bird singing in a tree?”



Lesson Plan: Listening Playground App

First Recording Together (10 minutes)

Teacher models the full process:

Press Record (7 seconds will record by default)

Take a photo of the sound source

Name the sound (e.g. 'Classroom Door')

Choose tags (e.g. footsteps, wind)

Save to the Gallery

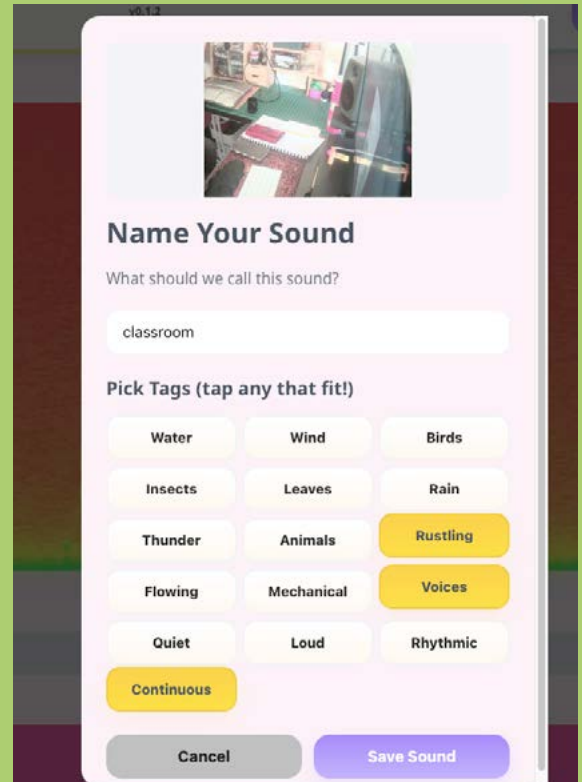
Teacher explains:

Naming and tagging sounds introduces the idea of organising collected data.

Tags describe qualities or textures - there is no single 'right' answer

You can collect badges for the kinds of sounds you can find, the app will reward you if you can find these sounds:

Quiet
Loud
Steady
Repeating
High
Low



Listening Exercise (5 minutes)

Eyes closed, 30 seconds of listening

Very close sounds

"First, listen for sounds very close to your body.

Can you hear your breathing?

Can you hear your clothes moving as you shift or breathe?

Maybe your tummy rumbling or your feet on the floor."

Nearby sounds

"Now listen for sounds just next to you.

A chair moving, someone breathing nearby"

Sounds in the room

"Now listen to the whole room.

Are there electronic hums, movement, or machines?"

Sounds just outside

"Now listen beyond the room.

What can you hear through the walls, doors, or windows?"

Faraway sounds

"Finally, listen for sounds that feel far away.

Traffic, wind, footsteps, or sounds you can't quite place."

Allow 30 seconds of quiet listening at the end.

Student prompts:

"Can anyone describe a sound that they heard?"

"Which sounds were close or far away?"

"If this sound were a shape, what shape would it be?"



Lesson Plan: Listening Playground App

Sound Hunt: Exploring the Environment (20 minutes)

Teacher puts the class into pre-selected groups and hands out the tablets ready to go

Instructions:

- Explore your assigned indoor or outdoor area
- Listen carefully and choose a small number of interesting or meaningful sounds
- Aim to collect a variety of sounds (remember the badges!)
- Focus on environmental and material sounds, rather than our own voices
- Take a photo of where each sound was found
- Name the sound and add tags to describe how the sound behaves or feels
- Use the sonogram to help guide your listening and recording choices

Teacher circulates as students explore to support, encourage non-voice environmental sounds and prompt slower, more attentive listening



Lesson Plan: Listening Playground App

Sharing What We Found: The Gallery (10 minutes)

Bring class together and point out the Pause All button at the top of the app.

Discussion prompts:

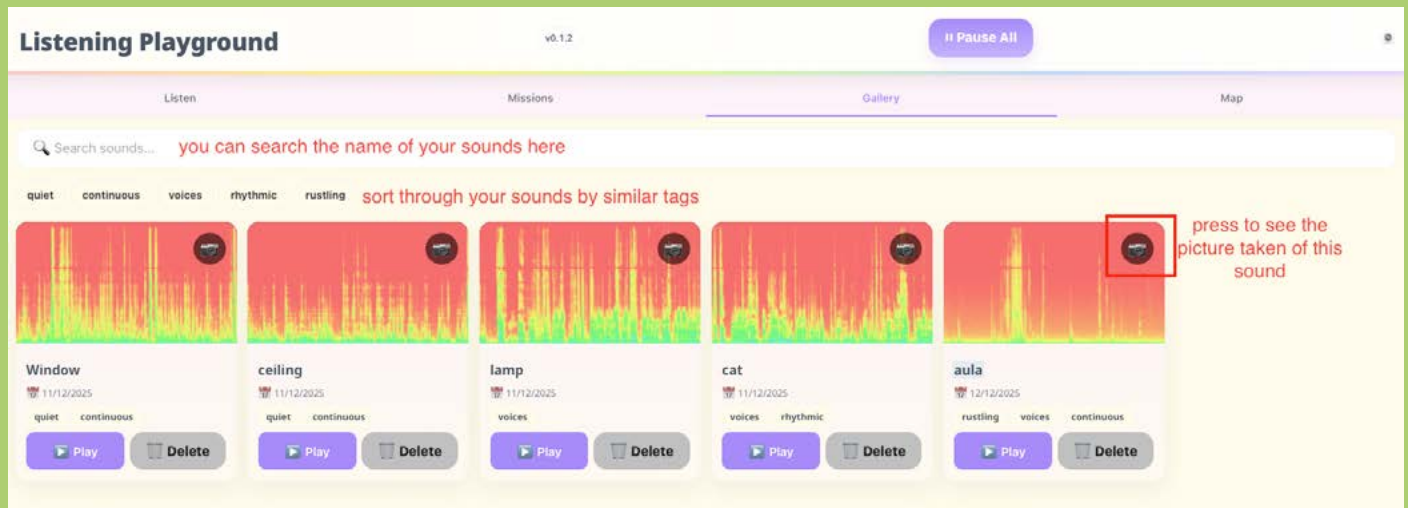
“What sounds did you find that surprised you?”

“Can we guess what kind of material made that sound?”

“Do any sounds look similar even if they sound different?”

“What kind of gain do you think this sound needed?”

“What do you think the sonogram looked like, and why?”



Making a Sound Map (10 minutes)

Teacher introduces the Map view:

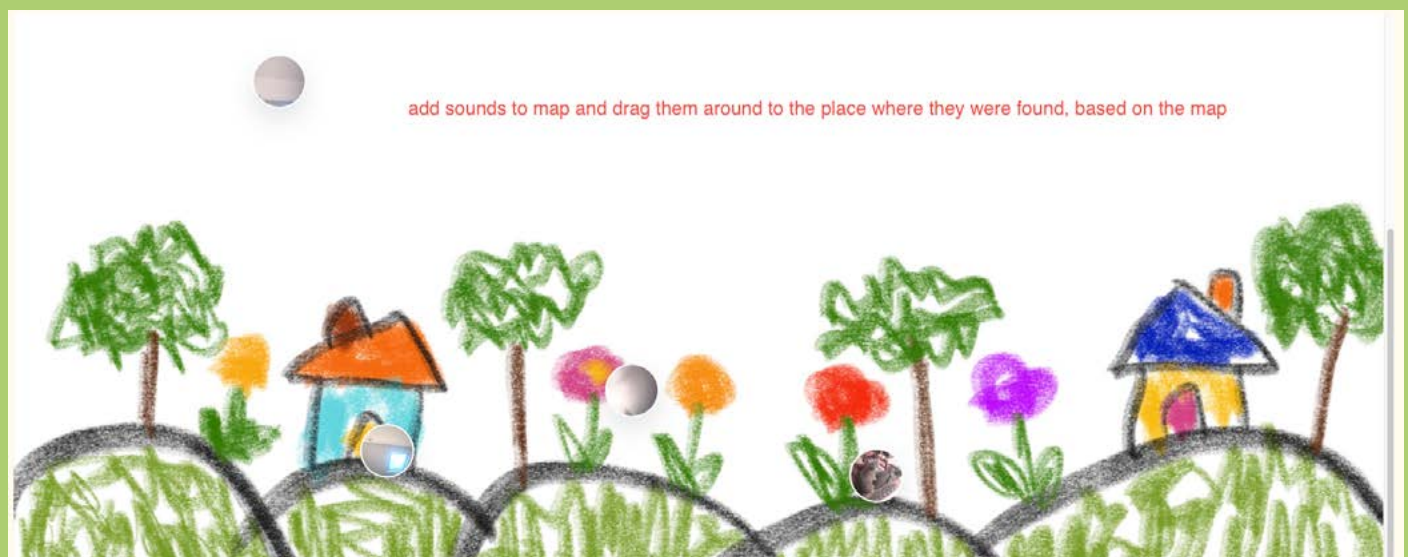
“Now we’re going to place our sounds on the map where we found them.”

Students:

Add sounds from top of the map by pressing the + button

Use photos, names, and tags in the gallery view to help remember locations

Teacher explains: the tags, names, and pictures you took can help you remember much about a sound that you have collected, where you found it, and how you might put it on the map.



Lesson Plan: Listening Playground App

Reflection & Wrap-Up (5 minutes)

Student reflection:

Favourite sound

Hardest sound to find

Did the sonogram look how you expected?

Teacher closes:

“You’ve created a sound portrait of your place.”

Explain that the map can grow over time or change with seasons.

Saving the Project (Teacher Guidance)

After the lesson, teachers can save all student work for later use or reflection.

To save the sound map:

1. Open Teacher Settings in the Sound Explorer app
2. Select Download Project Map

This downloads a ZIP file containing everything created during the session.

The downloaded folder includes:

- Map images: The background map used (drawing or photo)
- Sound recordings: Each recording saved as a WAV audio file and a matching text file containing:

- sound name
- tags

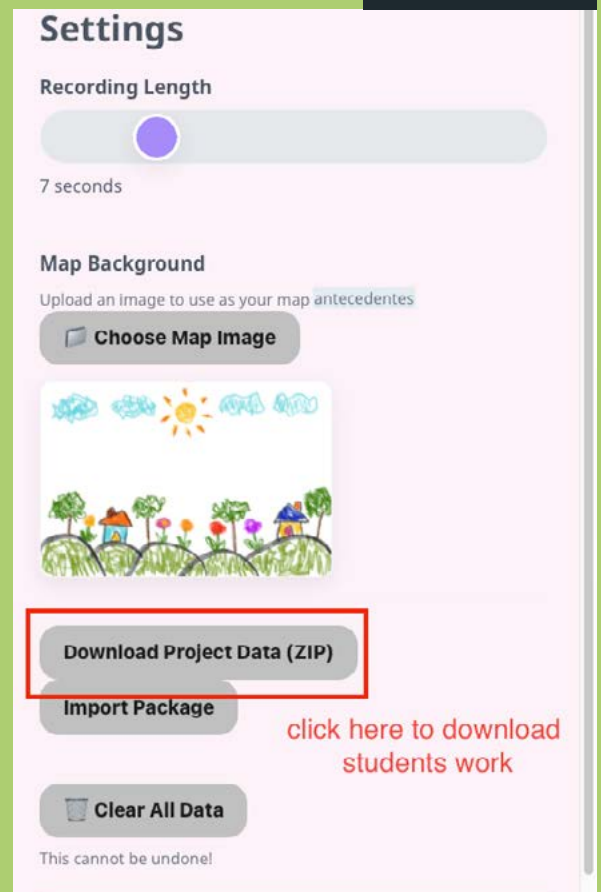
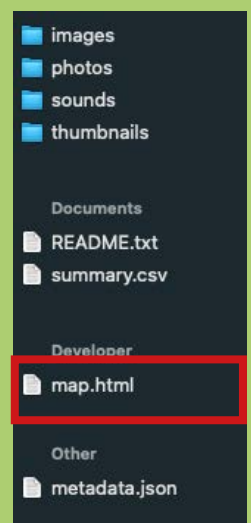
- Photos: All images students took of sound locations
- Sonogram thumbnails: A small image of the sonogram for each sound

- Summary spreadsheet: A list of all sounds and their associated data

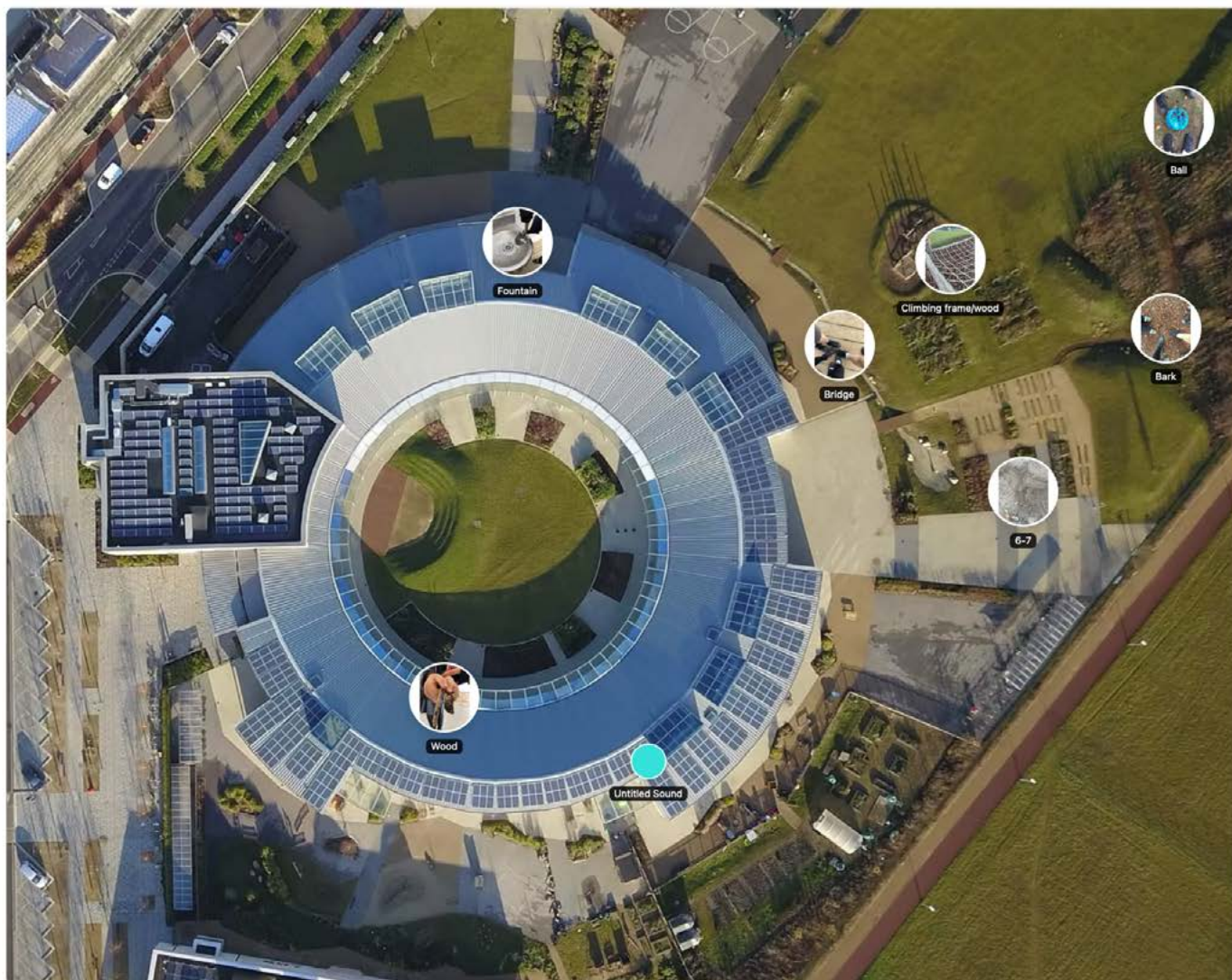
- HTML project file: An interactive version of the sound map

Revisiting the Map Later

- The HTML file is the easiest way to revisit the project
- Simply double-click the file to reopen the full sound map exactly as students left it
- This allows teachers and students to:
 - listen again to recordings
 - see and share how sounds were mapped
 - reflect on patterns and changes if future maps are made over time.



Listening Playground

[Hide Pictures](#)[Play All](#)[Download CSV](#)

Fountain

15/10/2025, 09:56:42 • 7s

water

Tags: water

[Play](#) [Download](#) [Tags Note](#)

Bridge

15/10/2025, 09:58:44 • 7s

rhythmic

Tags: rhythmic

[Play](#) [Download](#) [Tags Note](#)

Bark

15/10/2025, 10:00:24 • 7s

rustling

Tags: rustling

[Play](#) [Download](#) [Tags Note](#)

6-7

15/10/2025, 10:03:04 • 7s

rustling

Tags: rustling

[Play](#) [Download](#) [Tags Note](#)

Climbing frame/wood

15/10/2025, 10:04:22 • 7s

loud continuous

Tags: loud, continuous

[Play](#) [Download](#) [Tags Note](#)

Ball

15/10/2025, 10:07:31 • 7s

quiet

Tags: quiet

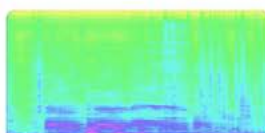
[Play](#) [Download](#) [Tags Note](#)

Wood

15/10/2025, 10:13:42 • 7s

quiet

Tags: quiet

[Play](#) [Download](#) [Tags Note](#)

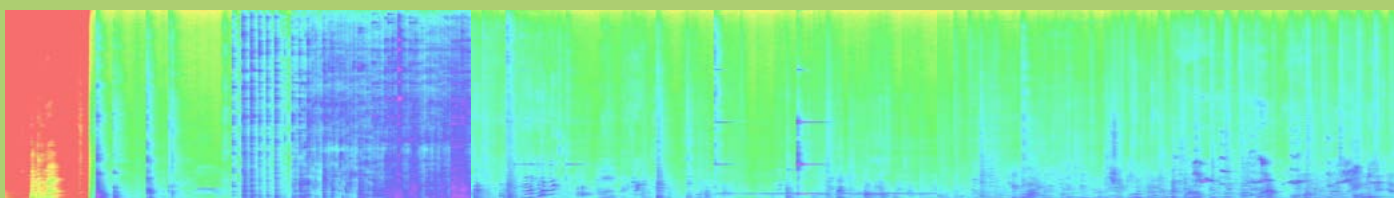
Untitled Sound

15/10/2025, 10:45:17 • 7s

No tags

[Play](#) [Download](#) [Tags Note](#)

Example saved map made by year 6 students at the University of Cambridge Primary School. Most photos shared in this document were taken by students themselves during their interaction with this activity.



Optional Extensions & Follow-On Activities

These activities can be used as short follow-ups, homework tasks, or as part of a longer project. With a focus on cross-curriculum learning.

Art: Sonogram Drawing

Invite students to draw or paint a sonogram they found interesting.

Focus on shape, colour, and movement thinking about the sonograms students have already made with the app.

Display drawings alongside sound titles or photos

Outcome: Visual artwork responding to sound

Writing: Sound Postcards

Students choose one sound and write a short descriptive paragraph or 'postcard'.

Encourage sensory language (what does it remind you of? how does it move? Think about the naming and tagging from the app.)

Outcome: Short creative writing piece

Science: Sound & Distance

Re-record the same sound from different distances.

Compare how the sonogram changes.

Discuss what this shows about sound energy and space.

Outcome: Simple observations and broader science discussions.

Community: Building a Sound Atlas

Invite students to record sounds from home or their neighbourhood.

Add them to the class sound map over time.

Compare how different places sound.

Outcome: Growing class sound archive

Composition: Making a Sound Piece

Students choose a small set of recorded sounds.

Decide an order to play them back (can be arranged in the map section with a different idea of a map as a composition).

Discuss whether/how this sequence can be like a piece of music.

Outcome: Short sound composition or listening piece

Performance: Sound as Action

Encourage students to record performative or musical sounds.

Explore how movement, gesture, or rhythm changes sound.

Outcome: Sound-based performance exploration



Listening Playground



Get in Touch & Future Projects

Digital Playgrounds for Music is an ongoing research and practice initiative exploring how listening-led, creative technologies can support musical learning, sensory inquiry, and cross-curricular creativity in schools and community settings.

If you use the Listening Playground app or lesson materials, we would love to hear how it works in your context.

DPfM@hud.ac.uk